

An explicit formula for lucky parking spots

A proof of the polynomial-form conjecture,
with reflection identities and boundary asymptotics

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Abstract

Let $C_{n,j}$ count ordinary parking functions on n spots in which spot j is occupied by a car that preferred that spot. We prove the explicit formula

$$C_{n,j} = \frac{j+1}{2j}(n+1)^{n-1} - \frac{1}{2j} \sum_{k=0}^{j-2} \binom{n}{k} j^k (n-j+1)^{n-1-k}, \quad 1 \leq j \leq n,$$

with the sum empty when $j = 1$. Factoring the sum proves Conjecture 13 of Butler, Hadaway, Lenius, Martens, and Moats, including the assertion that the correction polynomial has degree exactly $j - 2$. The leading coefficient is $\frac{1}{2j} \sum_{k=0}^{j-2} j^k / k!$. The proof uses a last-arrival decomposition, a bivariate exponential generating function, and two explicit applications of Lagrange inversion.

The formula also gives an exact reflection law, formulas for every fixed distance from the right endpoint, and in particular

$$C_{n,n-1} = \frac{(n+1)^{n-1} + (5n-1)(n-1)^{n-2}}{4}.$$

A binomial-probability interpretation gives both edge limits in terms of Poisson tails and proves convergence to $1/2$ whenever both distances to the ends tend to infinity. Further consequences include a reciprocity identity for the correction polynomials, a convergent fixed-column expansion in $1/n$, and an exact central-column formula. All proofs are given in ordinary mathematical language. The accompanying standard-library Python program performs exact, independently organized checks and regenerates the data.

Keywords: parking functions; lucky spots; tree function; Lagrange inversion; binomial tails; OEIS A374756 and A374533.

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1 The problem and the result

1.1 What is being counted?

Cars $1, \dots, n$ arrive in this order at a one-way street with spots $1, \dots, n$. Car i has a preference $a_i \in \{1, \dots, n\}$ and parks at the first unoccupied spot weakly to the right of a_i . A preference list is a *parking function* when all cars park successfully. A spot is *lucky* when the car occupying it preferred that very spot.

Write

$$P_n = (n + 1)^{n-1} \quad (n \geq 1), \quad P_0 = 1,$$

and let $C_{n,j}$ be the number of parking functions with lucky spot j . Thus $p_{n,j} = C_{n,j}/P_n$ is a probability under the uniform distribution on parking functions. The index j denotes a *spot*, not the arrival label of a car. These two statistics are different.

Butler et al. [1] give formulas for the first five columns and conjecture a polynomial form for every fixed column (Conjecture 13, p. 10). Their Question 15 asks about columns at fixed distance from the right end. The column triangle and the next-to-last column are associated with OEIS A374756 and A374533, respectively [2, 3].

1.2 An explicit answer

Theorem 1.1 (Lucky-spot formula). *For every pair of integers $1 \leq j \leq n$, put $x = n - j + 1$. Then*

$$C_{n,j} = \frac{j+1}{2j} P_n - \frac{1}{2j} \sum_{k=0}^{j-2} \binom{n}{k} j^k x^{n-1-k}. \quad (1.1)$$

The empty-sum convention makes this valid for $j = 1$ as well. For $j \geq 2$, define the rational polynomial

$$f_j(n) = \frac{1}{2j} \sum_{k=0}^{j-2} \binom{n}{k} j^k (n - j + 1)^{j-2-k}, \quad (1.2)$$

where $\binom{n}{k} = (n)_k/k!$ is interpreted polynomially. Then

$$C_{n,j} = \frac{j+1}{2j} (n+1)^{n-1} - f_j(n) (n-j+1)^{n-j+1}, \quad n \geq j, \quad (1.3)$$

and f_j has degree exactly $j - 2$, with leading coefficient

$$r_j = \frac{1}{2j} \sum_{k=0}^{j-2} \frac{j^k}{k!} > 0. \quad (1.4)$$

The exact degree is important: merely bounding it above would not prove the full conjecture. Here it follows immediately from (1.2), because every summand has degree $j - 2$ and positive leading coefficient. The remaining work is to establish (1.1) without assuming the conjectured form. Sections 2–4 provide that proof.

For orientation, the first polynomials are

$$\begin{aligned} f_2(n) &= \frac{1}{4}, \\ f_3(n) &= \frac{2n-1}{3}, \\ f_4(n) &= \frac{13n^2-26n+9}{8}, \\ f_5(n) &= \frac{(2n-3)(59n^2-177n+64)}{30}, \\ f_6(n) &= \frac{115n^4-920n^3+2436n^2-2384n+625}{12}. \end{aligned}$$

The cases through $j = 5$ agree with the previously published formulas [1]; the single expression (1.2) produces every column. For instance, it gives $C_{10,6} = 1\,258\,181\,726$.

1.3 Scope and provenance

The consulted journal version is in volume 29 (2026), article 26.1.1; its arXiv predecessor is 2412.07873, first submitted in December 2024. In those versions the target is explicitly a conjecture. Targeted searches made on 19 September 2026 did not locate a published resolution. This is a statement about the consulted sources, not a guarantee that no independent proof exists elsewhere.

This manuscript supplies a complete proposed proof and exact checks, rather than treating a finite data fit as a solution. The local-outcome counting method is established machinery; related fixed-lucky-car enumeration is discussed by Harris and Martinez [4]. No novelty is claimed for Lagrange inversion, the tree function, or the basic outcome factorization. The claimed contribution is the explicit *lucky-spot* formula and the consequences proved below. The proof has not been peer-reviewed or checked in a proof assistant.

2 A decomposition at the last arriving car

2.1 Preferences compatible with a fixed outcome

An *outcome permutation* is the permutation $\sigma = (\sigma_1, \dots, \sigma_n)$ in which car σ_i occupies spot i . For each i , define

$$b_i = \max(\{k < i : \sigma_k > \sigma_i\} \cup \{0\}), \quad \ell_i = i - b_i. \quad (2.1)$$

Thus b_i is the nearest spot to the left whose eventual occupant arrives *after* car σ_i .

Lemma 2.1 (Outcome factorization). *For a fixed outcome σ , the possible preference of car σ_i is any integer in $\{b_i + 1, \dots, i\}$, independently for each i . Consequently*

$$P_n = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n \ell_i, \quad C_{n,j} = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i \neq j} \ell_i. \quad (2.2)$$

Proof. Immediately before car σ_i arrives, every spot strictly between b_i and i is occupied: its occupant has smaller arrival label. Spot i is unoccupied, and if $b_i > 0$ then spot b_i is also unoccupied. A preference at most b_i cannot lead to spot i , because the car would encounter an unoccupied spot no later than b_i . A preference greater than i cannot lead backwards to i . Every preference between $b_i + 1$ and i does lead to i .

To justify independence, assign one permitted preference to each car and run the cars in increasing arrival-label order. Inductively, all earlier cars occupy their prescribed spots. The preceding argument then puts the current car in its prescribed spot too. Hence the choices really are independent and exhaust all preference lists for this outcome.

There are ℓ_i choices at spot i . Requiring spot j to be lucky fixes its car's preference to j , replacing that one factor by 1 and leaving all other factors unchanged. $\square$

For example, $\sigma = (2, 4, 1, 3)$ gives $(\ell_1, \ell_2, \ell_3, \ell_4) = (1, 2, 1, 2)$. There are four compatible parking functions; two have lucky spot 2, and two have lucky spot 4. This small example also illustrates why simply counting outcome permutations would lose the required weights.

2.2 Splitting the street

Suppose the last arriving car, labeled n , occupies spot $a + 1$. There are a spots to its left and $b = n - 1 - a$ to its right. Choose the left labels in $\binom{n-1}{a}$ ways. After standardizing labels within each side, the left and right outcomes are arbitrary outcomes of lengths a and b .

The weight of the last car is $a + 1$, because all preceding spots are already occupied. For a car on the right, the last car supplies a permanent not-yet-arrived boundary while that right-side car parks; subtracting $a + 1$ from spot numbers gives exactly the ordinary right-subproblem lengths. Thus there is no hidden interaction across the split.

Define a polynomial marking the *location* of one lucky spot:

$$H_n(z) = \sum_{j=1}^n C_{n,j} z^{j-1}, \quad H_0(z) = 0. \quad (2.3)$$

This is a sum of indicators over possible marked spots, not the generating polynomial of the total number of lucky spots.

Proposition 2.2 (Last-arrival recurrence). *For $n \geq 1$,*

$$H_n(z) = \sum_{a+b=n-1} \binom{n-1}{a} ((a+1)H_a(z)P_b + (a+1)z^{a+1}P_a H_b(z) + z^a P_a P_b). \quad (2.4)$$

Proof. The marked spot is on the left, on the right, or is occupied by car n . In the first two cases the last car has its unrestricted factor $a + 1$. A right-side marked spot is shifted by $a + 1$ positions, giving z^{a+1} . In the third case car n itself is required to be lucky, so its factor is 1, and its spot contributes z^a . Lemma 2.1 supplies the remaining independent factors. $\square$

The recurrence is already an exact algorithm for the entire triangle. It will also provide a computational check that is independent of the final binomial expression.

3 The bivariate generating function

3.1 The tree function and Lagrange inversion

Let $T = T(t)$ be the unique formal power series with zero constant term satisfying

$$T = te^T. \quad (3.1)$$

The Lagrange inversion formula in the form needed here is

$$[t^m]F(T(t)) = \frac{1}{m}[u^{m-1}]F'(u)e^{mu}, \quad m \geq 1. \quad (3.2)$$

For completeness, this follows by substituting $t = ue^{-u}$ in the formal residue for the coefficient. The resulting residue is $\text{Res } F(u)(1-u)e^{mu}u^{-m-1}du$; the residue of the derivative of $F(u)e^{mu}u^{-m}$ is zero, yielding (3.2).

In particular,

$$[t^m]e^{cT(t)} = \frac{c(m+c)^{m-1}}{m!} \quad (m \geq 1), \quad T(t) = \sum_{m \geq 1} \tau_m t^m, \quad \tau_m = \frac{m^{m-1}}{m!}. \quad (3.3)$$

This identity applies to positive or negative c ; no analytic convergence is needed for its formal use. The familiar total $P_n = (n+1)^{n-1}$ can be counted by arranging $n+1$ spots on a circle. All $(n+1)^n$ preference lists park, leaving one spot empty. Rotating every preference rotates the empty spot, and every orbit has exactly one list leaving spot $n+1$ empty. Such a list is precisely an ordinary parking function on the first n spots: no car can prefer or traverse the spot that remains empty. Therefore

$$P(t) := \sum_{n \geq 0} P_n \frac{t^n}{n!} = e^{T(t)}, \quad T = tP(t), \quad T'(t) = \frac{P(t)}{1-T(t)}. \quad (3.4)$$

The coefficient identity also follows directly from (3.3) with $c = 1$.

3.2 A linear differential equation

Set

$$H(z, t) = \sum_{n \geq 0} H_n(z) \frac{t^n}{n!}.$$

Multiplication of exponential generating functions absorbs the binomial coefficient in (2.4). Its three terms give

$$H_t = (H + tH_t)P(t) + z(P(z) + ztP'(zt))H + P(t)P(zt).$$

Equivalently,

$$(1 - T(t))H_t = (P(t) + \frac{d}{dt}T(zt))H + P(t)P(zt). \quad (3.5)$$

This is linear in H and, with $H(z, 0) = 0$, determines it uniquely.

Make the change of variables

$$u = T(t), \quad v = T(zt), \quad t = ue^{-u}.$$

Throughout a derivative in u holds z fixed. The identity $ve^{-v} = zue^{-u}$ gives

$$\frac{dv}{du} = \frac{v(1-u)}{u(1-v)}, \quad e^v = \frac{ve^u}{zu}. \quad (3.6)$$

Equation (3.5) becomes

$$H_u = \left(1 + \frac{v}{u(1-v)}\right)H + e^v. \quad (3.7)$$

A homogeneous solution is

$$L(u, z) = e^u \frac{u - v}{u}, \quad \frac{L_u}{L} = 1 + \frac{v}{u(1 - v)}.$$

Variation of constants therefore gives

$$H(z, t) = \frac{e^u(u - v)}{zu} \int_0^u \frac{T(ze^{-s})}{s - T(ze^{-s})} ds. \quad (3.8)$$

All divisions in this expression have a harmless formal interpretation. The series $T(ze^{-s})/s$ is well-defined and divisible by z , so $1 - T(ze^{-s})/s$ is invertible as a series in z . The integral is also divisible by z , canceling the displayed $1/z$. Finally $(u - v)/u$ is well-defined and has constant term $1 - z$ in u . In particular, no assertion about a singular analytic integral is being smuggled into the proof.

4 Extracting the exact formula

4.1 The first coefficient extraction

For $j \geq 1$, define

$$Q_j(s) = \sum_{h=0}^{j-1} \left(1 - \frac{h}{j}\right) \frac{(js)^h}{h!}, \quad (4.1)$$

$$R_j(u) = \sum_{h=0}^{j-1} \frac{(j-h)(j-h+1)}{2j^2} \frac{(ju)^h}{h!}, \quad c_j = \frac{j+1}{2j}. \quad (4.2)$$

Lagrange inversion in the variable z gives

$$\begin{aligned} [z^j] \frac{T(ze^{-s})}{s - T(ze^{-s})} &= \frac{(se^{-s})^j}{j} [w^{j-1}] \frac{se^{jw}}{(s-w)^2} \\ &= e^{-js} Q_j(s). \end{aligned} \quad (4.3)$$

The last equality follows by expanding $(1 - w/s)^{-2}$; the apparent negative powers of s cancel against s^j .

A direct differentiation of the polynomial (4.2) shows that $R'_j = jR_j - Q_j$ and $R_j(0) = c_j$. Hence, coefficientwise,

$$I_j(u) := \int_0^u e^{-js} Q_j(s) ds = c_j - e^{-ju} R_j(u). \quad (4.4)$$

Define

$$S_j(u) = \sum_{a=1}^{j-1} \tau_a u^{a-1} R_{j-a}(u) - R_j(u). \quad (4.5)$$

Since $v = \sum_{a \geq 1} \tau_a z^a u^a e^{-au}$, extracting z^{j-1} from (3.8) yields

$$[z^{j-1}] H(z, t) = c_j e^u - \sum_{a=1}^{j-1} c_{j-a} \tau_a t^{a-1} + e^{-(j-1)u} S_j(u). \quad (4.6)$$

The fact that $u^{a-1} e^{-(a-1)u} = t^{a-1}$ is essential here: the middle expression is a finite polynomial in t , not an additional infinite series contributing to large n .

4.2 The convolution that makes the formula simple

Lemma 4.1 (Closed form of the auxiliary polynomial). *For $j \geq 2$,*

$$S_j(u) = \frac{1}{2j} \sum_{h=0}^{j-2} \frac{(ju)^h}{h!(j-h-1)} - \frac{j^{j-3}}{(j-1)!} u^{j-1}. \quad (4.7)$$

Also $S_1(u) = -1$.

Proof. The degree of each summand in the convolution in (4.5) is at most $j-2$. Thus the coefficient of u^{j-1} comes only from $-R_j$ and equals the last coefficient in (4.7). The constant term, for $j \geq 2$, is

$$R_{j-1}(0) - R_j(0) = \frac{j}{2(j-1)} - \frac{j+1}{2j} = \frac{1}{2j(j-1)}.$$

It remains to treat $1 \leq h \leq j-2$.

Put $m = h+1$ and $c = j-m \geq 1$. Expanding (4.5), the convolution part of the coefficient of u^h is

$$\frac{c(c+1)}{2} \sum_{a=1}^m \frac{a^{a-1}}{a!} \frac{(j-a)^{m-a-2}}{(m-a)!}. \quad (4.8)$$

Introduce the formal series

$$U_c(t) = \sum_{b \geq 0} \frac{(b+c)^{b-2}}{b!} t^b.$$

Multiplying the coefficient at t^b by $b+c$ and using (3.3) shows that

$$tU'_c(t) + cU_c(t) = \frac{1}{c} e^{cT(t)}.$$

Its constant term is $1/c^2$, and direct differentiation gives the unique solution

$$U_c(t) = e^{cT(t)} \left(\frac{1}{c^2} - \frac{T(t)}{c(c+1)} \right). \quad (4.9)$$

The sum in (4.8) is $[t^m]T(t)U_c(t)$. Two applications of (3.2), with $m+c=j$, give

$$\begin{aligned} [t^m]T(t)e^{cT(t)} &= \frac{(c+1)j^{m-2}}{(m-1)!}, \\ [t^m]T(t)^2e^{cT(t)} &= \frac{(c+2)j^{m-3}}{(m-2)!}. \end{aligned}$$

The second identity is being used only for $m \geq 2$. Therefore the sum in (4.8) equals

$$\frac{(c+1)j^{m-2}}{c^2(m-1)!} - \frac{(c+2)j^{m-3}}{c(c+1)(m-2)!}.$$

After multiplication by $c(c+1)/2$ and subtraction of the coefficient of R_j , the coefficient of S_j is

$$\begin{aligned} \frac{j^{m-3}}{2(m-1)!} \left(\frac{(c+1)^2 j}{c} - (c+2)(m-1) - (c+1)(c+2) \right) &= \frac{j^{m-2}}{2c(m-1)!} \\ &= \frac{j^{h-1}}{2h!(j-h-1)}. \end{aligned}$$

In the simplification we used $j = m+c$. This is exactly the coefficient specified by (4.7). Finally $S_1 = -R_1 = -1$ follows directly. $\square$

4.3 The final extraction and proof of the conjecture

Proof of Theorem 1.1. For $j = 1$, equation (4.6) reads $[z^0]H = e^{T(t)} - 1$, so $C_{n,1} = P_n$ for $n \geq 1$. Suppose $j \geq 2$. In (4.6), the middle polynomial has degree at most $j - 2$. The final term of (4.7), after multiplication by $e^{-(j-1)u}$, is a constant multiple of $u^{j-1}e^{-(j-1)u} = t^{j-1}$. Neither contributes to $[t^n]$ for $n \geq j$.

The remaining terms, besides $c_j e^{T(t)}$, are

$$\frac{1}{2j} \sum_{h=0}^{j-2} \frac{j^h}{h!(j-h-1)} t^h e^{-(j-h-1)T(t)}.$$

Let $b = j - h - 1 > 0$ and $m = n - h \geq 1$. By (3.3),

$$[t^m] e^{-bT(t)} = -\frac{b(m-b)^{m-1}}{m!}.$$

Here $m - b = n - j + 1 = x \geq 1$. Multiplying by $n!$ shows that the contribution of the term indexed by h is

$$-\frac{1}{2j} \binom{n}{h} j^h x^{n-1-h}.$$

Adding these contributions proves (1.1). Factoring x^{n-j+1} gives (1.2) and (1.3). Each term in (1.2) has degree $j - 2$ and leading coefficient $j^k/(2j k!)$. Their sum is the strictly positive number (1.4), proving the exact degree and completing the proof. $\square$

Remark 4.2 (Why finite interpolation would have been insufficient). Computing many values of $C_{n,j}$ and fitting a polynomial after dividing by x^{n-j+1} would only test the conjecture. The proof instead explains why the infinite generating function collapses to the particular finite binomial sum: the unwanted terms become polynomials in t supported strictly below $n = j$. This cancellation is the structural step, and Lemma 4.1 makes it explicit.

5 Binomial probabilities, reflection, and the right endpoint

5.1 An exact probability representation

Corollary 5.1 (Binomial-tail representation). *Let $x = n - j + 1$ and $Y \sim \text{Bin}(n, j/(n+1))$. Then*

$$p_{n,j} = \frac{j+1}{2j} - \frac{n+1}{2jx} \mathbb{P}(Y \leq j-2). \quad (5.1)$$

Equivalently,

$$p_{n,j} = \frac{1}{2} + \frac{1}{2j} \mathbb{P}(Y \geq j-1) - \frac{1}{2x} \mathbb{P}(Y \leq j-2). \quad (5.2)$$

Proof. Since $j + x = n + 1$,

$$\mathbb{P}(Y \leq j-2) = \frac{1}{(n+1)^n} \sum_{k=0}^{j-2} \binom{n}{k} j^k x^{n-k}.$$

Divide (1.1) by P_n and use $(n+1)/(jx) = 1/j + 1/x$. The same calculation works for $j = 1$ with $\mathbb{P}(Y \leq -1) = 0$. $\square$

This binomial random variable is a representation of the final answer; the proof does not assume that preference choices remain independent after conditioning on successful parking.

Theorem 5.2 (Reflection identity). *For $1 \leq j \leq n$ and $x = n - j + 1$,*

$$C_{n,j} + C_{n,x} = P_n + \frac{n!}{j!x!} j^{j-1} x^{x-1}. \quad (5.3)$$

Proof. For $Y \sim \text{Bin}(n, j/(n+1))$, the corresponding binomial variable for column x has the distribution of $n - Y$. Thus the two tail probabilities appearing in (5.1) are $\mathbb{P}(Y \leq j - 2)$ and $\mathbb{P}(Y \geq j + 1)$. Their sum is $1 - \mathbb{P}(Y = j - 1) - \mathbb{P}(Y = j)$.

The two missing masses are equal, because

$$\frac{\mathbb{P}(Y = j)}{\mathbb{P}(Y = j - 1)} = \frac{n - j + 1}{j} \frac{j}{x} = 1.$$

The constant parts in the two formulas sum to $1 + (n + 1)/(2jx)$. The tail coefficients are both $(n + 1)/(2jx)$. Consequently

$$p_{n,j} + p_{n,x} - 1 = \frac{n + 1}{jx} \binom{n}{j} \frac{j^j x^{n-j}}{(n + 1)^n}.$$

Multiplication by P_n and the identity $x! = (n - j + 1)(n - j)!$ give (5.3). $\square$

The extra term has a direct relation to the last car. The number of parking functions in which car n is lucky at spot j is

$$L_{n,j} = \binom{n-1}{j-1} P_{j-1} P_{n-j};$$

this follows by splitting the street into the two completely filled sides and fixing its final preference to j . Thus (5.3) is also $C_{n,j} + C_{n,n-j+1} = P_n + nL_{n,j}$.

5.2 All fixed right-edge columns

Corollary 5.3 (Right-edge polynomial form). *For $j \geq 2$, set*

$$g_j(n) = (n - j + 1)f_j(n) + \frac{j^{j-1}}{j!} (n)_{\underline{j-1}}. \quad (5.4)$$

For every $n \geq j$,

$$C_{n,n-j+1} = \frac{j-1}{2j} P_n + g_j(n)(n - j + 1)^{n-j}. \quad (5.5)$$

The polynomial g_j has degree $j - 1$, with leading coefficient

$$r_j + \tau_j = \frac{1}{2j} \sum_{k=0}^j \frac{j^k}{k!}. \quad (5.6)$$

For $j = 1$, take $f_1 = 0$ and $g_1 = 1$, obtaining $C_{n,n} = n^{n-1}$.

Proof. Subtract (1.3) from (5.3) and factor x^{n-j} . This gives (5.4)–(5.5). Both leading coefficients in (5.4) are positive. Finally the terms with indices $j-1$ and j in the sum in (5.6) are equal and each is $j\tau_j$ before division by $2j$. $\square$

In particular,

$$C_{n,n-1} = \frac{(n+1)^{n-1} + (5n-1)(n-1)^{n-2}}{4}, \quad n \geq 2, \quad (5.7)$$

$$C_{n,n-2} = \frac{1}{3}(n+1)^{n-1} + \frac{13n^2 - 19n + 4}{6}(n-2)^{n-3}, \quad n \geq 3. \quad (5.8)$$

Equation (5.7) gives a closed form for A374533, whose nine listed values at the time of consultation are reproduced by the formula [3]. More generally, (5.4) answers the fixed-distance version of Question 15 by finite polynomial arithmetic, not by a recurrence whose order grows with n .

5.3 The central column and a row-sum check

Corollary 5.4 (Exact central probability). *For $m \geq 1$,*

$$\frac{C_{2m-1,m}}{(2m)^{2m-2}} = \frac{1}{2} + \frac{1}{m4^m} \binom{2m}{m}. \quad (5.9)$$

Proof. Set $n = 2m - 1$ and $j = x = m$ in (5.3), then divide by twice P_n . The remaining factorials simplify to the central binomial coefficient in (5.9). $\square$

Summing (5.3) over j also gives a useful global check. From the last-car decomposition and (3.4),

$$\sum_{j=1}^n L_{n,j} = (n-1)! [t^{n-1}] P(t)^2 = 2(n+1)^{n-2}.$$

The coefficient statement at $n = 1$ is read directly as 1; the displayed closed expression also equals 1 there. Hence

$$\sum_{j=1}^n C_{n,j} = P_n \frac{n(n+3)}{2(n+1)}. \quad (5.10)$$

This counts marked lucky spots, not parking functions themselves, and is consistent with the known expected lucky-car count because each lucky car occupies exactly one lucky spot.

6 Structure of the correction polynomials

The expression (1.2) is positive for integers $n \geq j$, even though its expansion in ordinary powers of n has alternating signs in the first examples. In particular,

$$0 < C_{n,j} < \frac{j+1}{2j} P_n \quad (2 \leq j \leq n).$$

Positivity of $C_{n,j}$ also follows directly by taking any outcome with all preferences equal to their final spots.

There is a second, less obvious structure. Put $m = j - 2$. The polynomial formula is equivalent to

$$f_j(n) = \frac{1}{2j} [z^m] \frac{(1 + jz)^n}{1 - (n - j + 1)z}. \quad (6.1)$$

Here $(1 + jz)^n$ is expanded by generalized binomial coefficients, so the identity makes sense with n an indeterminate.

Proposition 6.1 (Polynomial reciprocity). *For $j \geq 2$,*

$$f_j(j - 2 - n) = (-1)^{j-2} f_j(n). \quad (6.2)$$

Thus $f_j(y + (j - 2)/2)$ contains only powers of y with the same parity as $j - 2$. For odd j , $2n - j + 2$ is a factor of $f_j(n)$.

Proof. Write (6.1) as a formal residue and substitute $z = -w/(1 + jw)$. This substitution has nonzero linear coefficient. Since $m = j - 2$ and $x = n - j + 1$, it gives

$$\begin{aligned} 2j f_j(n) &= \text{Res}_{z=0} \frac{(1 + jz)^n}{1 - xz} \frac{dz}{z^{m+1}} \\ &= (-1)^m \text{Res}_{w=0} \frac{(1 + jw)^{m-n}}{1 + (n + 1)w} \frac{dw}{w^{m+1}} \\ &= (-1)^m 2j f_j(m - n). \end{aligned}$$

The final denominator is exactly the one associated with $n' = m - n$, since $n' - j + 1 = -n - 1$. Centering the variable gives the parity statement, and an odd polynomial has a zero at its center. $\square$

For example,

$$\begin{aligned} f_4(y + 1) &= \frac{13y^2 - 4}{8}, \\ f_5(y + 3/2) &= \frac{y(236y^2 - 275)}{60}, \\ f_6(y + 2) &= \frac{115y^4 - 324y^2 + 81}{12}. \end{aligned}$$

This explains the linear factors in the odd-column examples rather than merely recording them.

For $j \geq 3$, the coefficient of n^{j-3} follows from the center of symmetry:

$$f_j(n) = r_j n^{j-2} - \frac{(j-2)^2}{2} r_j n^{j-3} + O(n^{j-4}). \quad (6.3)$$

For $j = 2$, use $f_2 = r_2 = 1/4$ instead. No assertion about the other zeros, such as their reality, is needed or made here.

7 Boundary limits, bulk behavior, and expansions

7.1 Fixed distance from either edge

Theorem 7.1 (The two edge limits). *For each fixed $j \geq 1$, let $Z_j \sim \text{Pois}(j)$. Then*

$$\rho_j := \lim_{n \rightarrow \infty} p_{n,j} = \frac{j+1}{2j} - \frac{e^{-j}}{2j} \sum_{k=0}^{j-2} \frac{j^k}{k!} = \frac{1}{2} + \frac{1}{2j} \mathbb{P}(Z_j \geq j-1), \quad (7.1)$$

$$\beta_{j-1} := \lim_{n \rightarrow \infty} p_{n,n-j+1} = \frac{j-1}{2j} + \frac{e^{-j}}{2j} \sum_{k=0}^j \frac{j^k}{k!} = \frac{1}{2} - \frac{1}{2j} \mathbb{P}(Z_j \geq j+1). \quad (7.2)$$

In particular, $\rho_j > 1/2$ and $\beta_{j-1} < 1/2$ for every finite j .

Proof. For fixed j , a binomial random variable with parameters n and $j/(n+1)$ converges to $\text{Pois}(j)$. At each fixed nonnegative integer k , this follows directly from

$$\binom{n}{k} \left(\frac{j}{n+1} \right)^k \left(1 - \frac{j}{n+1} \right)^{n-k} \rightarrow e^{-j} \frac{j^k}{k!}.$$

Only finitely many such masses occur in (5.1), proving (7.1).

Alternatively using the reflection identity, its extra term divided by P_n tends to $\tau_j e^{-j}$. Hence the right limit equals $1 - \rho_j + \tau_j e^{-j}$. Equation (5.6) gives the stated finite sum. The probability forms and strict inequalities follow because both Poisson tails have positive probability. $\square$

Table 1: Limiting probabilities at fixed distance from the two endpoints. The right column uses spot $n - j + 1$, not spot j . Values are computed from the proved finite sums.

j	r_j	ρ_j (left)	β_{j-1} (right)
1	0	1.0000000000	0.3678794412
2	1/4	0.7161661792	0.4191691040
3	2/3	0.6334752878	0.4412053148
4	13/8	0.5952370868	0.4536046169
5	59/15	0.5734974085	0.4615960655
6	115/12	0.5595786250	0.4671918985
8	42037/720	0.5429141077	0.4745342088
10	461843/1260	0.5333590161	0.4791519875

In particular,

$$\beta_0 = e^{-1}, \quad \beta_1 = \frac{1}{4} + \frac{5}{4}e^{-2}, \quad \beta_2 = \frac{1}{3} + \frac{13}{6}e^{-3}.$$

As $j \rightarrow \infty$, the standardized variable $(Z_j - j)/\sqrt{j}$ tends to a standard normal variable. One direct verification expands its characteristic function as $\exp\{j(e^{it/\sqrt{j}} - 1 - it/\sqrt{j})\} \rightarrow e^{-t^2/2}$. The thresholds $j-1$ and $j+1$ are both within $O(1)$ of the mean, so

$$\rho_j = \frac{1}{2} + \frac{1}{4j} + o(j^{-1}), \quad \beta_{j-1} = \frac{1}{2} - \frac{1}{4j} + o(j^{-1}). \quad (7.3)$$

These are iterated edge limits: first $n \rightarrow \infty$ with j fixed, then $j \rightarrow \infty$. The next subsection treats moving columns directly.

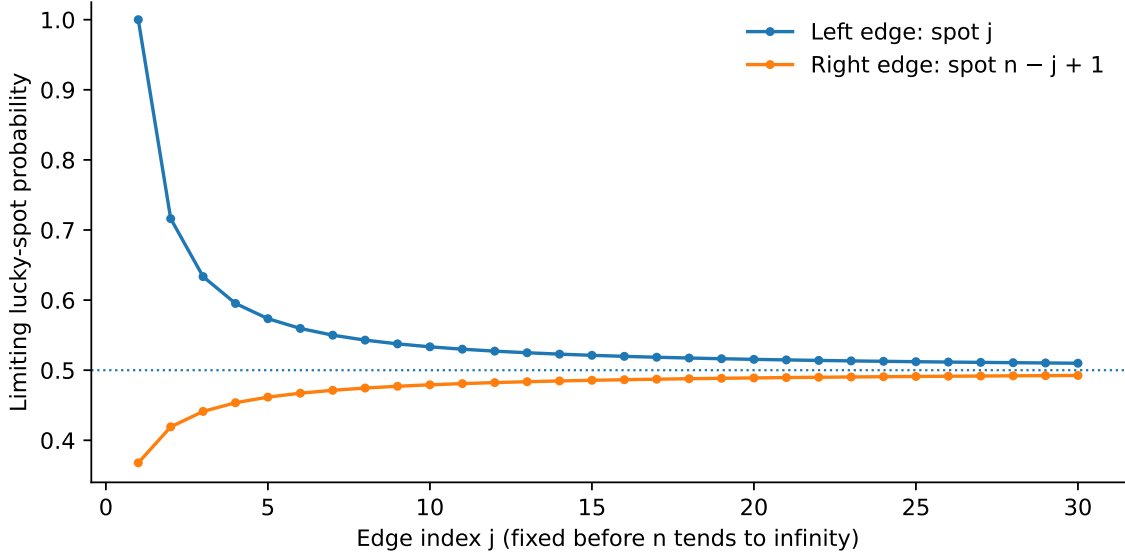


Figure 1: The two fixed-edge limits from Theorem 7.1. The plot illustrates the proved formulas; it is not used to infer monotonicity.

7.2 A uniform bulk limit

Corollary 7.2 (All columns escaping both endpoints). *For every $1 \leq j \leq n$, with $x = n - j + 1$,*

$$\frac{1}{2} - \frac{1}{2x} \leq p_{n,j} \leq \frac{1}{2} + \frac{1}{2j}. \quad (7.4)$$

Consequently, for any sequence of column positions $j = j(n)$ such that $j(n) \rightarrow \infty$ and $n - j(n) + 1 \rightarrow \infty$, one has $p_{n,j(n)} \rightarrow 1/2$.

Proof. Both tail probabilities in (5.2) lie in $[0, 1]$. The displayed bounds follow immediately and their errors tend to zero under the stated hypotheses. $\square$

No relation between the two rates of escape is required. For instance, $j = \lfloor \log n \rfloor$ is covered, not just columns proportional to n . If more specifically $j/n \rightarrow \alpha \in (0, 1)$, the binomial central limit theorem gives $\mathbb{P}(Y \leq j - 2) \rightarrow 1/2$: the threshold differs from the mean by a bounded amount, whereas the standard deviation is of order $\sqrt{n}$. Substitution in (5.1) then yields the sharper limit

$$n \left(p_{n,j} - \frac{1}{2} \right) \rightarrow \frac{1 - 2\alpha}{4\alpha(1 - \alpha)}. \quad (7.5)$$

For $\alpha = 1/2$ this first correction vanishes, and the exactly centered odd-size column has a different leading scale. From (5.9) and Stirling's expansion,

$$p_{2m-1,m} = \frac{1}{2} + \frac{1}{\sqrt{\pi} m^{3/2}} \left(1 - \frac{1}{8m} + \frac{1}{128m^2} + O(m^{-3}) \right). \quad (7.6)$$

For clarity, this expansion follows by inserting $\log(m!) = (m + \frac{1}{2}) \log m - m + \frac{1}{2} \log(2\pi) + 1/(12m) + O(m^{-3})$ into $(2m)!/(m!)^2$ and exponentiating. It is not obtained by substituting a moving j into a fixed-column expansion.

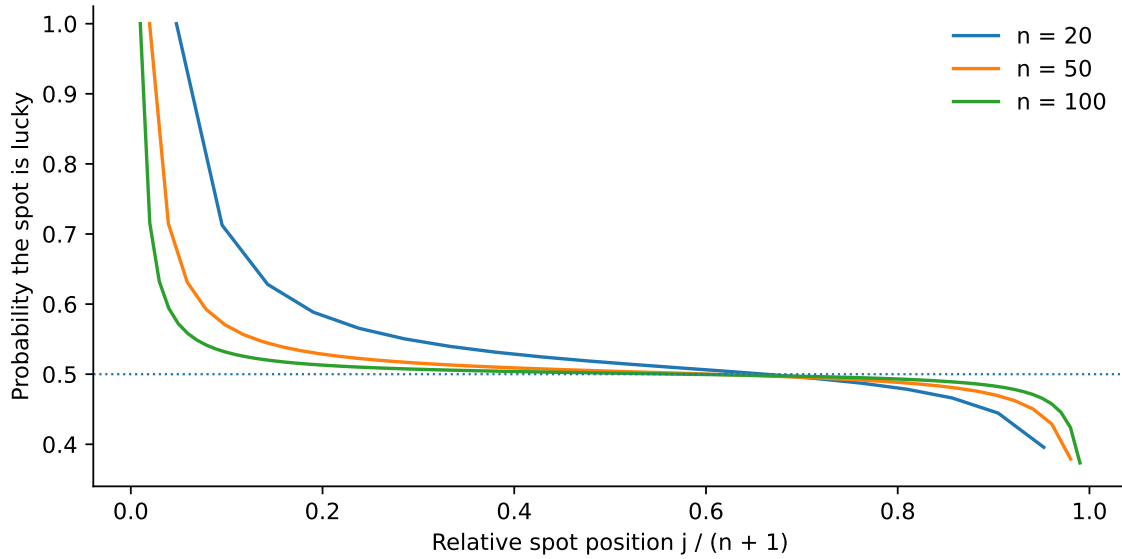


Figure 2: Exact lucky-spot probabilities at three street lengths. The near-end boundary regions coexist with convergence to $1/2$ in the interior. All plotted counts were computed by exact integer arithmetic before conversion to plotting coordinates.

7.3 A convergent fixed-column expansion

Fix $j \geq 2$ and write $d = j - 1$. An exact logarithmic expansion for $n > d$ is

$$\log \left(n^{j-2} \frac{(n-d)^{n-d}}{(n+1)^{n-1}} \right) = -j + \sum_{q \geq 1} \frac{d^{q+1} - (-1)^q (2q+1)}{q(q+1)n^q}. \quad (7.7)$$

To obtain it, expand $(n-d) \log(1-d/n)$ and $(n-1) \log(1+1/n)$ and subtract. Both logarithmic series converge under the stated inequality, so this is a convergent expansion, not just a formal asymptotic rule.

Together with the finite polynomial $f_j(n)/n^{j-2}$, equation (7.7) provides every coefficient of the expansion of $p_{n,j}$ in powers of $1/n$. Using (6.3), the first correction simplifies to

$$p_{n,j} = \rho_j - \frac{j r_j e^{-j}}{n} + O(n^{-2}). \quad (7.8)$$

Indeed, the exponential contributes $(d^2 + 3)/(2n)$, and the polynomial contributes $-(j-2)^2/(2n)$. Their sum is j/n . The same calculation with the constant polynomial handles $j = 2$.

The positive coefficient shows that, for each fixed $j \geq 2$, the probabilities approach their limiting value from below once n is large enough. This does not by itself prove monotonicity in n , and no such stronger assertion is required here.

8 Exact computations and reproducibility

8.1 Independent routes to the same counts

The archive includes `code/verify.py`, which uses only the Python standard library. Its count comparisons use integers and rational numbers; no floating-point tolerances enter any pass/fail

decision. It compares four independently organized enumerations:

1. Direct simulation of every preference list in $\{1, \dots, n\}^n$. A failed car rejects the list; successful cars directly record which spots were lucky.
2. A dynamic program on the occupied-spot bitmask. For every state it tries each possible next preference and accumulates both the number of histories and the vector of lucky-spot counts.
3. Enumeration of outcome permutations with the factors (2.1), checking Lemma 2.1 against direct dynamics.
4. The last-arrival recurrence (2.4), compared against the closed expression (1.1).

The first two methods use the parking process itself; they do not assume the outcome-weight lemma or the final formula. The recurrence and the closed form are distinct algorithms, although the proof explains their connection.

Table 2: Executed verification coverage. Every listed check passed.

Check	Executed range
Closed formula versus last-arrival recurrence	all $1 \leq j \leq n \leq 100$; 5 050 pairs
Reflection identity	the same 5 050 pairs
Exhaustive preference-list simulation	$n \leq 7$; 873 612 lists
Direct occupied-state dynamic program	$n \leq 12$
Weighted outcome permutations	$n \leq 8$; 46 233 permutations
Polynomial coefficients, exact degree, parity, and Lemma 4.1	$2 \leq j \leq 30$
Published table prefixes and A374533	10 row prefixes; 9 terms
First/last columns, row sums, next-to-last formula, central formula	all applicable $n \leq 100$

A machine-readable transcript in `data/verification.json` records these bounds and the actual interpreter version. The run used Python 3.13.5 and completed in approximately two seconds in the working environment; timing is incidental, not a benchmark claim.

8.2 Minimal exact evaluator

The final formula requires very little code. This evaluator uses only integer arithmetic and explicitly checks divisibility:

```
from math import comb

def lucky_spot_count(n: int, j: int) -> int:
    if not 1 <= j <= n:
        raise ValueError("Expected 1 <= j <= n")
    x = n - j + 1
    correction = sum(
        comb(n, k) * j**k * x**(n - 1 - k)
        for k in range(j - 1)
    )
    numerator = (j + 1) * (n + 1)**(n - 1) - correction
    value, remainder = divmod(numerator, 2 * j)
    if remainder:
        raise ArithmeticError("Unexpected nonintegral count")
    return value
```

For a single count this has $j - 1$ summands, with additional cost for large-integer arithmetic and exponentiation. Reflection permits evaluation through the nearer endpoint. It is important not to describe such a bound as constant-time arithmetic for arbitrarily large n : the output itself has $\Theta(n \log n)$ bits.

For all rows through N , the straightforward last-arrival recurrence uses $O(N^3)$ arithmetic operations and $O(N^2)$ stored integers. These are arithmetic-operation bounds; bit complexity grows with the integer sizes. The explicit formula is especially convenient for a fixed column or a fixed distance from the right edge.

8.3 Archive contents and commands

The package contains the complete L^AT_EX source and compiled PDF; `verify.py`; optional plotting code; CSV data for all 5 050 entries through $n = 100$; exact rational polynomial coefficients through $j = 30$; 50-digit edge-limit tables; and a local extension of A374533. The figure PDFs are vector graphics. No external fonts or copies of the cited papers are redistributed.

From the package directory, run

```
python code/verify.py
python code/make_figures.py      # optional; requires matplotlib
pdflatex -interaction=nonstopmode article.tex
pdflatex -interaction=nonstopmode article.tex
```

The first command needs Python 3.9 or later and no third-party packages. The verification bounds can be changed with command-line options described by `python code/verify.py -help`. A normal T_EX Live or MiK_TE_X installation with the packages named in the source can compile the article. The supplied figures mean plotting is not necessary to rebuild the PDF.

8.4 An OEIS indexing caution

The consulted A374756 entry labels its statistic as a lucky *car* in the headline, while its comment and displayed triangle specify a lucky *spot*. Its longer example also reproduces only the first six columns

of the later rows. Our comparison therefore uses the stated (n, j) coordinates and the published row prefixes, not an assumed flattening of an incomplete triangle. The generated CSV supplies the complete triangular array through $n = 100$. A374533 has offset $n = 2$ and is unambiguous [2, 3].

9 Conclusion and remaining directions

The selected target admits an explicit solution. The key result is not merely the existence of a correction polynomial but the formula

$$f_j(n) = \frac{1}{2j} \sum_{k=0}^{j-2} \binom{n}{k} j^k (n-j+1)^{j-2-k}.$$

It proves the exact degree, identifies the formerly unspecified rational coefficient in the limiting probability, and reveals binomial tails behind the column counts. The reflection identity then transfers fixed-column information to the opposite endpoint, including the next-to-last-column question, without a separate case analysis.

The resulting asymptotic picture is also precise. Fixed left-edge columns have limiting probability greater than $1/2$; fixed right-edge columns have limiting probability less than $1/2$; and every sequence of columns escaping both ends converges to $1/2$. At the exact center, the first nonzero correction has scale $n^{-3/2}$ rather than n^{-1} .

There are natural extensions, but they are not prerequisites for the results proved here. A direct bijective explanation of the binomial tail or the reflection identity would complement the generating-function proof. Marking several specified lucky spots would lead to joint probabilities and correlations; the present one-mark recurrence does not already solve that problem. Finally, monotonicity questions and a uniform higher-order transition expansion between edge and bulk regimes merit separate arguments. Numerical patterns alone should not be promoted to theorems.

The mathematical conclusions of this manuscript rest on the displayed proofs. The accompanying finite checks are intended to expose indexing, weighting, and algebraic errors, and to make independent examination easy. They are not substituted for those proofs.

References

- [1] S. Butler, K. Hadaway, V. Lenius, P. Martens, and M. Moats, *Lucky Cars and Lucky Spots in Parking Functions*, Journal of Integer Sequences **29** (2026), Article 26.1.1. Conjecture 13 and Question 15. <https://cs.uwaterloo.ca/journals/JIS/VOL29/Hadaway/had3.pdf>. Preprint: arXiv:2412.07873 (2024), <https://arxiv.org/abs/2412.07873>.
- [2] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A374756, lucky-spot triangle (see the entry's comment and example). <https://oeis.org/A374756>. Accessed 19 September 2026.
- [3] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A374533, parking functions whose next-to-last spot is lucky. <https://oeis.org/A374533>. Accessed 19 September 2026.
- [4] P. E. Harris and L. Martinez, *Parking functions with a fixed set of lucky cars*, arXiv:2410.08057v1 (2024), especially the outcome-counting formulation in Theorem 2.6. <https://arxiv.org/abs/2410.08057>.